



WASDCAT Games
INDEPENDENT GAME STUDIO

Low Poly Colorizer

Official User Manual

Compact guide for streamlined low-poly asset coloring in
Blender and optimized material and texture export for
Godot 4.

VERSION

1.0.0

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Foreword

Low Poly Colorizer (LPC) is a Blender extension designed for fast, intuitive, and resource-efficient color and material styling of 3D models in low-poly aesthetics. Developed for game production at **WASDCAT Games**, it bridges the gap between rapid 3D prototyping in Blender and optimal draw-call performance in modern game engines such as **Godot 4**.

The LPC Principle: References Instead of Material Chaos

Traditional low-poly workflows often struggle with two extremes: either the number of material slots explodes through dozens of individual materials, or static albedo texture atlases prohibit flexible physical properties (PBR).

Low Poly Colorizer solves this through a consistent **reference architecture**:

- **Color Reference (Albedo):** Each face stores discrete coordinates (X, Y) referencing a dynamically generated palette texture (`lpc_palette.png`).
- **Material Look (PBR Preset):** Each face references a named preset (*Solid, Metallic, Emission, Clearcoat*), resolved via a compact lookup table (`lpc_preset_lut.png`).
- **A Single Material:** All painted objects share the `lpc_multicolor` material. In the game engine, each object requires **only a single draw call**.
- **Live Propagation:** Adjusting palette saturation or preset roughness later updates **all painted surfaces across the entire project instantly**, without re-traversing meshes.

Publication Notice

This user manual was typeset and published using the modern Markdown publishing tool [markpublish](#).

1 Installation and Quickstart

1.1 Online Installation via Extensions (Recommended)

Starting with Blender 4.2, Low Poly Colorizer can be installed directly through Blender's extension repository:

1. In Blender, navigate to **Edit** ▶ **Preferences** ▶ **Get Extensions**.
2. Type **Low Poly Colorizer** into the search bar and click **Install**. The extension is automatically downloaded, installed, and registered in the N-Panel.

1.2 Manual Installation (ZIP Archive)

For offline workstations or pre-release builds:

1. In Blender, navigate to **Edit** ▶ **Preferences** ▶ **Get Extensions** (or *Add-ons*).
2. Click the menu icon (▼) in the top right corner and select **Install from Disk....**
3. Select the downloaded `low-poly-colorizer-x.y.z.zip` archive. The sidebar panel (**N-Panel** ▶ **LPC**) is immediately available.

1.3 Quickstart in 60 Seconds

Once the addon is enabled, you can start painting right away:

1. **Prepare a model:** Select a mesh object in the 3D Viewport (e.g. the default cube). Optionally press `Tab` to switch to **Edit Mode** and select individual faces.
2. **Open the N-Panel:** Press `N` in the 3D Viewport and switch to the **LPC** tab.
3. **Select a preset:** In the preset list, *Solid* is active by default.
4. **Open the palette:** Click the **Palette button** (color picker) next to the coordinates. The GPU palette overlay opens directly in the viewport.
5. **Assign color:** Click any color cell. The picker closes and the selection is painted in real time with the chosen color and material look.

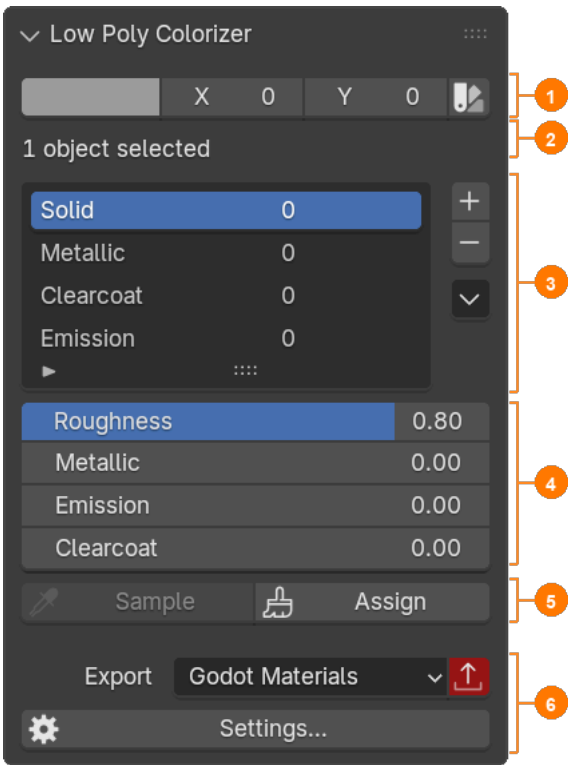
💡 Tip

Live Assignment: Clicking a cell in the modal picker performs an assignment automatically. Clicking the "Assign" button afterward is not necessary.

2 The User Interface

All controls in Low Poly Colorizer are organized within a compact sidebar panel in the 3D Viewport (**N-Panel** ▶ **LPC**).

2.1 The N-Panel at a Glance



No.	Section	Description
①	Color Selection & Picker	Displays the color and coordinates (X, Y) of the active palette cell. Clicking the Palette button (color picker) opens the interactive modal picker in the viewport. Coordinates can also be edited numerically.
②	Selection Status	Dynamically reports the target selection (" <i>14 faces in 1 object selected</i> " or " <i>1 object selected</i> "). Displays a warning if no preset is selected.

No.	Section	Description
③	Preset List & Actions	Lists all material presets. The number on the right indicates the refcount (how many faces currently use this preset). Includes buttons to add (+), delete (-), and access the actions menu (↗).
④	PBR Properties	Controls the physical material properties (<i>Roughness, Metallic, Emission, Clearcoat</i>) of the currently active preset . Changes propagate instantly to all faces using this preset.
⑤	Tools	Assign: Assigns the active brush (color + preset) to the selection. Sample: Eyedropper tool to pick color and preset from an existing face. Select / Deselect: (<i>Edit Mode only</i>) Selects or deselects all faces sharing identical color and preset.
⑥	Export & Settings	Target format selection (default: <i>Godot Materials</i>), Export button with automatic dirty tracking (glows red when unsaved changes exist), and button to open Settings....

2.2 Protection & Feedback Mechanisms

- **Refcount Protection:** A preset cannot be deleted while its reference counter is greater than 0, preventing accidental broken mesh references.
- **Visual Dirty Tracking:** Whenever presets, colors, or face assignments change, the Export button highlights in vibrant **red**. After a successful export, it returns to neutral.
- **Color & Preset Matching:** *Select* and *Deselect* use strict logical AND matching: only faces that share **both the exact palette cell and the same preset** are affected.

2.3 Shortcuts & Workflow Hotkeys

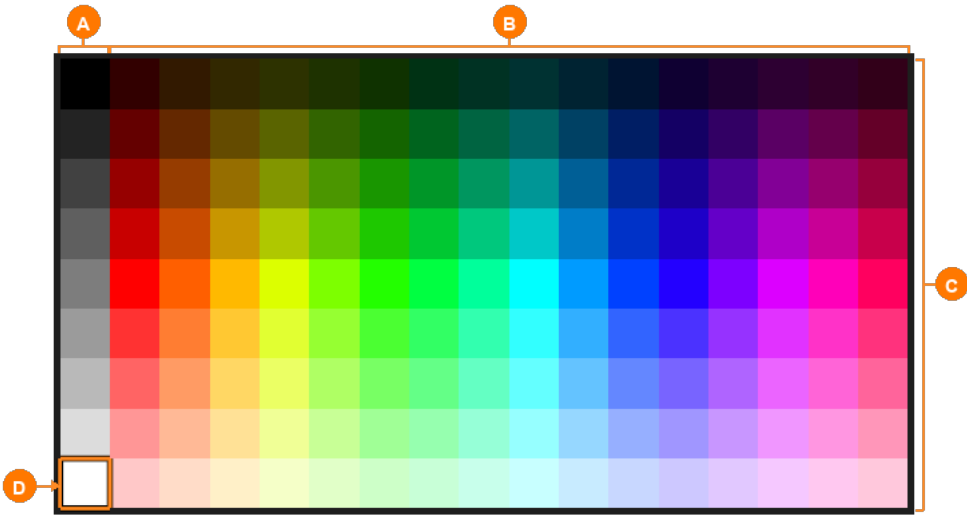
Key	Context	Function
N	3D Viewport	Toggle sidebar containing the LPC panel
Tab	3D Viewport	Toggle between Object Mode (paint whole object) and Edit Mode
1 / 2 / 3	Edit Mode	Switch between Vertex, Edge, and Face selection modes
L	Edit Mode (cursor over face)	Select linked geometry island
A / Alt + A	3D Viewport	Select all / Deselect all

3 Color Palette and PBR Presets

In Low Poly Colorizer, color values (albedo) and surface characteristics (PBR parameters) are managed independently and combined during rendering and shading.

3.1 The Interactive Modal Palette Picker

Clicking the **Palette button** (color picker) in the N-Panel opens the color palette as a hardware-accelerated GPU overlay directly within the 3D Viewport.



Area	Description
Ⓐ	Greyscale Column (Column 0): Neutral greys from pure black (top) to pure white (bottom) for monochrome surfaces or neutral shading.
Ⓑ	Hue Spectrum (Columns 1–16): Full HSV color wheel from red through yellow, green, cyan, blue, to magenta laid out horizontally.
Ⓒ	Brightness & Saturation Gradient (Rows 0–7): Vertical progression from deep shadow tones (top) through rich saturated colors to pale pastels (bottom).
Ⓓ	Active Cell & Selection Frame: Two-tone highlight frame indicating the currently selected cell (X,Y) with instant assignment on click.

3.1.1 Navigation within the Picker Overlay

- **Select & Assign:** Click with the **left mouse button (LMB)** on any cell. The coordinates are stored, assigned to the current selection, and the overlay closes.
- **Zoom:** Scroll the **mouse wheel** to zoom continuously into or out of the palette grid (0.25× to 4.0×).
- **Pan:** Hold the **middle mouse button (MMB)** and drag to reposition the palette window anywhere in the viewport.
- **Cancel:** Press **Right-Click**, **Escape**, or click outside the palette to close the overlay without changing the current color selection.

3.2 PBR Presets: Managing Material Looks

Each preset combines four standardized physical material parameters:

- **Roughness (0.0 – 1.0):** Surface micro-roughness. Lower values yield sharp mirror reflections; higher values produce a soft, diffuse matte look.
- **Metallic (0.0 – 1.0):** Metallic proportion. 0.0 corresponds to dielectrics (plastic, wood, stone); 1.0 corresponds to true conductive metals.
- **Emission (0.0 – 10.0):** Surface self-illumination strength. The cell's albedo color serves as the emission tint.
- **Clearcoat (0.0 – 1.0):** Secondary protective clear lacquer layer (e.g. automotive paints, varnished surfaces, or wet finishes).

Important

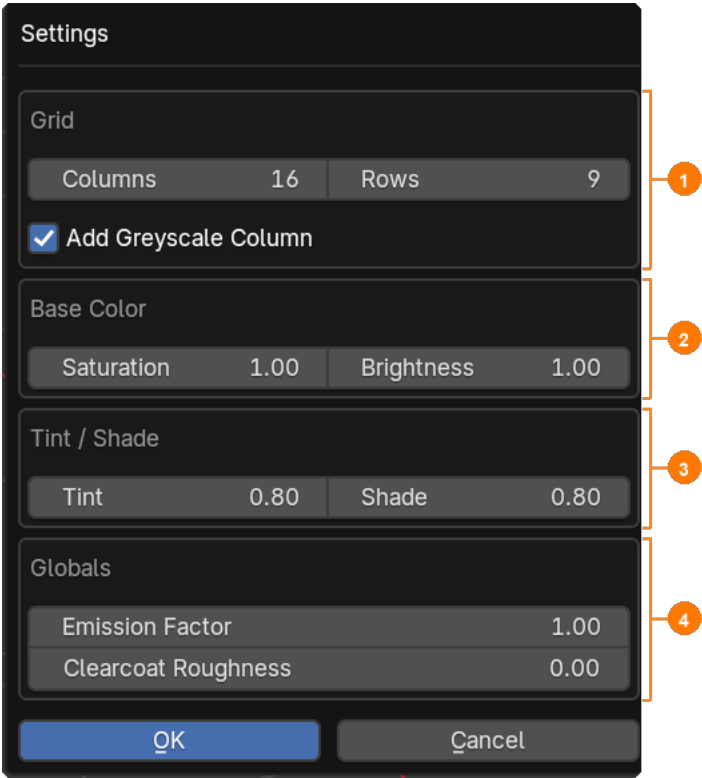
Live Scatter Architecture: Preset sliders do not modify individual mesh vertices or face loops; they update the central lookup table. Adjusting the *Roughness* of the *Solid* preset updates **every painted face in the entire project instantaneously**.

3.2.1 Extended Preset Menu (•)

- **Duplicate:** Creates a 1:1 duplicate of the active preset to quickly iterate on material variations.
- **Import / Export JSON:** Saves and loads preset definitions as portable JSON files—ideal for sharing material sets across projects.
- **Load Default Presets:** Restores the four bundled factory presets (*Solid*, *Metallic*, *Clearcoat*, *Emission*).

3.3 Global Settings (*Settings...*)

Clicking **Settings...** in the N-Panel opens the configuration dialog for project-wide palette generation and shader parameters:



Area	Description
①	Grid: Column (<i>Columns</i>) and row (<i>Rows</i>) counts for the palette grid, plus a toggle for the monochrome greyscale column (<i>Add Greyscale Column</i>).
②	Base Color: Base saturation (<i>Saturation</i>) and base brightness (<i>Brightness</i>) for all hues in the color wheel.
③	Tint / Shade: Strength of the vertical gradation steps (highlighting via <i>Tint</i> , darkening via <i>Shade</i>).
④	Globals: Project-wide parameters including the global emission multiplier (<i>Emission Factor</i>) and base clearcoat roughness (<i>Clearcoat Roughness</i>).

4 Engine Export and Integration

Low Poly Colorizer is engineered from the ground up for maximum rendering performance in real-time game engines. The export process translates Blender scene data into lightweight, optimized textures and shaders.

4.1 One-Click Export to Godot

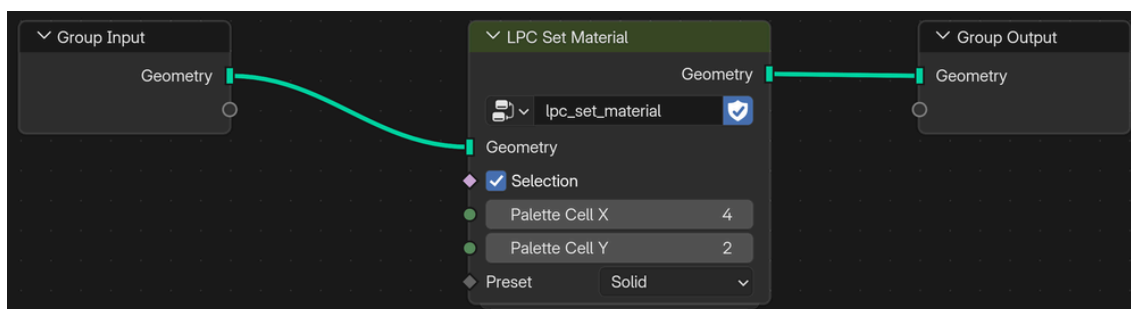
To prepare your assets for Godot, click the **Export button** in the N-Panel (next to the *Godot Materials* dropdown). LPC generates three files:

- **lpc_palette.png**: Palette texture storing all albedo and emission colors.
- **lpc_preset_lut.png**: Compact lookup table defining roughness, metallic, clearcoat, and emission parameters.
- **lpc_shader.tres**: Production-ready Godot shader that automatically binds both textures using dedicated UV sets.
- **Dirty Flag Tracking**: The export button illuminates in red whenever unsaved changes exist and resets to neutral once synchronized.

4.2 Procedural Workflows with Geometry Nodes

For procedural meshes, generate the `lpc_set_material` node group via the preset menu (▼) by selecting **Create Geometry Node Group**:

1. LPC constructs the node group and attaches it as a Geometry Nodes modifier to the active object.
2. In the Node Editor, drive palette coordinates (X, Y) and material preset procedurally through input sockets. The shared `lpc_multicolor` material handles shading automatically.



4.3 Maintenance: Repairing UV Maps (*Fix UV Maps*)

Low Poly Colorizer utilizes two designated UV channels: `lpc_uv0` (palette color coordinates) and `lpc_uv1` (preset LUT coordinates).

When meshes are joined (`Ctrl + J`), cut with Boolean operations, or imported from external sources, UV layers can become swapped or omitted. In this case, LPC automatically displays a highlighted warning box at the bottom of the N-Panel:

Warning

UV maps need fixing: Clicking **Fix UV Maps** repairs mesh attributes, restores the correct `lpc_uv0` / `lpc_uv1` layer ordering, and ensures seamless import into Godot.

4.4 Troubleshooting Checklist

Problem	Cause	Solution
No colors in viewport	3D Viewport is set to <i>Wireframe</i> or <i>Solid</i> mode without texture preview enabled.	Press <code>Z</code> and switch shading mode to Material Preview or Rendered .
Faces do not color	In Edit Mode, no faces are selected, or no preset is selected in the list.	Select faces with <code>A</code> or <code>L</code> and ensure a preset is highlighted in the list.
Minus button (-) disabled	The selected preset is currently used by mesh faces (refcount > 0).	Use <i>Select</i> to locate referencing faces and assign them a different preset.
Exported look differs in Godot	Textures were not re-exported after modifying presets (red export button).	Click the red Export button in the N-Panel to update texture assets.